COURSE SYLLABUS

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)	05.10.2020 08.10.2020	Monday Thursday
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS	12.10.2020 15.10.2020	Monday Thursday
3	TRAFFIC FLOW AND CAPACITY ROADWAY CLASSIFICATION AND GEOMETRIC STANDARDS	19.10.2020 22.10.2020	Monday Thursday
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**	26.10.2020 29.10.2020	Monday Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION	02.11.2020 05.11.2020	Monday Thursday
6	**APPLICATION (VEHICLE MOVEMENTS)** **ASSIGNMENT - APPLICATION (TRAFFIC)**	09.11.2020 12.11.2020	Monday Thursday
7	**APPLICATION (HORIZANTAL CURVE)** **ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)**	16.11.2020 19.11.2020	Monday Thursday
8	**MID TERM EXAM**	23.11.2020 26.11.2020	Monday Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES – ASSIGNMENT (RETURN)	30.11.2020 03.12.2020	Monday Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020 10.12.2020	Monday Thursday
11	**APPLICATION (TRANSITION CURVES)** **ASSIGNMENT**	14.12.2020 17.12.2020	Monday Thursday
12	**APPLICATION (VERTICAL CURVE)** **APPLICATION (VERTICAL AND HORIZANTAL CURVE)**	21.12.2020 24.12.2020	Monday Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)	28.12.2020 31.12.2020	Monday Thursday
14	DRAINAGE IN HIGHWAY **MAKEUP EXAM**	04.01.2020 07.01.2020	Monday Thursday

SELECTION OF GEOMETRIC STANDARDS

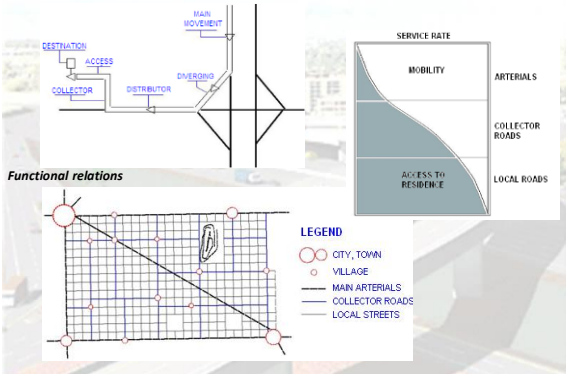
Geometrical standards of a highway are; platform width (width of lanes and banquette), minimum horizontal and vertical curve radii, maximum grades and slopes, (crown, cross slope or champer), bridge width, minimum underpass height, right of way.

Geometrical standards of a highway are selected based on project traffic and design speed, terrain type, type of highway (urban or rural) and the classification of highway are also the other important parameters.

Classification of highways

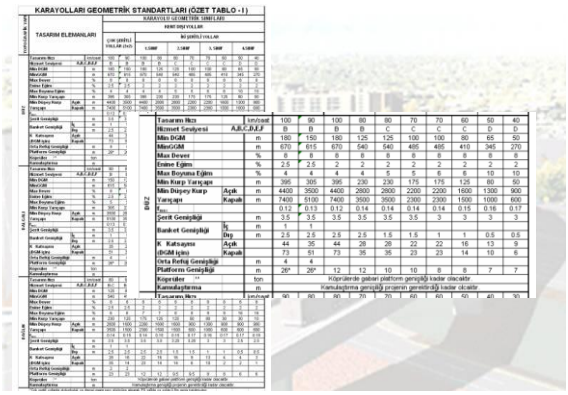
- According to the properties of the zone that highway passes from:
  - Urban highways: Highways surrounded by buildings, lighted, and having pedestrian traffic. They are divided into subcategories like artery (speeway), boulevard, avenue, street, alley.
  - Suburban highways: Suburban highway classes that are used on various countries are as follows:
    - 1st, 2nd, 3rd class highways.
    - Highway, expressway, speedway.
    - State highway, provincial road, rural road.
- According to the traffic value and geometry: Main road, side road.
- According to the vehicle class that uses the highway: Automobile road, pedestrian road, bicycle road, bus road.
- According to the type of the traffic: Residence road, travel road, work-commerce road.
- According to the situation of the highway platform: Divided road, undivided road.
- According to the flow direction of the traffic: One-sided road, two sided road.
- According to the type of the pavement: Asphalt road, concrete road, stabilized road, unpaved road, parquet road.

The hierarchy of the movement



Functional relations

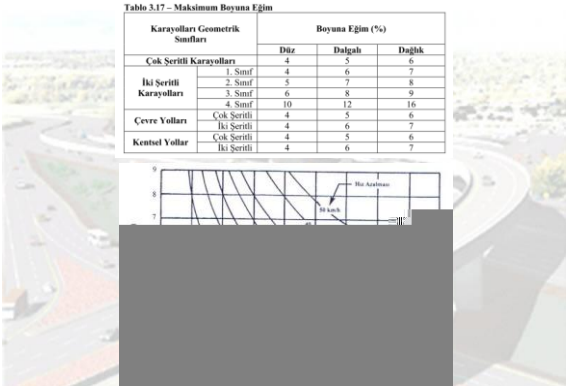
SELECTION OF GEOMETRIC STANDARDS



KARAYOLLARI GEOMETRİK STANDARTLARI (ÖZET TABLO - I)

TASARIM ELEMANLARI				KARAYOLU GEOMETRİK SINIFLARI										
				ANA YOL SINIFLARI										
				İKİ ŞERİTLİ YOLLAR										
				1. SINIF		2. SINIF		3. SINIF		4. SINIF				
				DAĞLIK		DALGALI		DÜZ		DÜZ				
Tasarım Hızı			km/saat	100	90	100	80	80	70	70	60	60	40	
Hizmet Seviyesi			A,B,C,D,E,F	B	B	B	B	C	C	C	C	D	D	
Min.DGM			m	180	150	180	125	125	100	100	80	65	50	
MinGGM			m	670	615	670	540	540	485	485	410	345	270	
Max.Devamlı			%	6	6	6	6	8	8	8	8	8	8	
Enine Eğim			%	2.5	2.5	2	2	2	2	2	2	2	2	
Max.Boyuna Eğim			%	4	4	4	4	5	5	6	6	10	10	
Min.Kısa Yarıçapı			m	395	305	395	230	230	175	175	125	80	50	
Min.Düşey Kısa		Açık	m	4400	3500	4400	2800	2800	2200	2200	1600	1300	900	
Yarıçapı		Kapalı	m	7400	5100	7400	3500	3500	2300	2300	1500	1000	600	
a_{min}				0.12	0.13	0.12	0.14	0.14	0.14	0.14	0.15	0.16	0.17	
Şerit Genişliği			m	3.5	3.5	3.5	3.5	3.5	3.5	3	3	3	3	
Banket Genişliği	İç		m	1										
	Dış		m	2.5	2.5	2.5	2.5	1.5	1.5	1	1	0.5	0.5	
K. Kahsayma		Açık		44	35	44	28	28	22	22	16	13	9	
(DGM için)		Kapalı		73	51	73	36	36	23	23	14	10	6	
Orta Refüj Genişliği			m	4	4									
Platform Genişliği			m	26*	26*	12	12	10	10	8	8	7	7	
Kaplamalar			tür	Köprülerde gabarı platform genişliği kadar olacaktır										
Kanalizasyonlama			m	Kanalizasyonlama genişliği projenin gerektirdiği kadar olacaktır										
Tasarım Hızı			km/saat	120	100	120	100	100	80	80	60	60	40	

SELECTION OF GEOMETRIC STANDARDS



Tablo 3.17 – Maksimum Boyuna Eğim

Karayolları Geometrik Sınıfları		Boyuna Eğim (%)		
		Düz	Dalgalı	Dağlık
Çok Şeritli Karayolları		4	5	6
İki Şeritli Karayolları	1. Sınıf	4	6	7
	2. Sınıf	5	7	8
	3. Sınıf	6	8	9
	4. Sınıf	10	12	16
Çevre Yolları	Çok Şeritli	4	5	6
	İki Şeritli	4	6	7
Kentsel Yollar	Çok Şeritli	4	5	6
	İki Şeritli	4	6	7

HIGHWAY ROUTE STUDY

The route is the direction that a highway follows on the land. There are many possibilities to connect two points through a highway. Selection of **the most appropriate route** is named as **the route study**.

Subjects that should be taken into account during the route study

- Route has to connect the main control points to each other.
- Route has to be propriate for the standards that the selected highway class requires.
- Route has to serve the traffic expected to pass from the highway during its project life.
- Route has to be suitable for the main purpose of the highway. It shall satisfy the desired social, economical, industrial, commercial, touristic purposes.
- Route has to pass from the geologically suitable places.
- Route has to be selected in a way to decrease the earthwork costs.
- Route has to pass from places without drainage problems.
- Route has to be perpendicular to river passes in order to decrease the bridge costs.
- Route has to pass from places that the material suppliance is easy.
- Route has to pass from places with low right of way costs.
- Route has to pass from places with low maintenance costs (north-south difference).

HIGHWAY ROUTE STUDY

Phases on Route Study

Preliminary Survey: On preliminary survey phase, the terrain that the highway is expected to pass is examined and potential feasible selections are determined.

Survey: A more detailed research on the feasible routes selected after the preliminary survey phase is followed on the survey phase. In this phase, topographical survey and soil surveys are made in general.

Economical Comparison: The last comparison made to determine the certain route is the economical comparison. Using the data gathered, information about the various highway cost items are obtained.

HIGHWAY ROUTE STUDY

Preliminary Study: On preliminary survey phase, the terrain that the highway is expected to pass is examined and potential feasible selections are determined. For this reason, topographical maps with a scale of 1/25.000 and geological maps with a scale of 1/100.000 are used. The feasible selections determined on the map are examined topographically, geologically and geotechnically on the terrain. The first eleminations are made accordingly.

Followings are examined during the study.

- The topographical situation of the terrain,
- The geological formation, geotechnical structure and landslide condition through the route,
- Condition of the underwater and surface waters and also the drainage possibilities,
- Situation of the potential material quarries,
- The place, type, approximate sizes and costs of potential engineering structures,
- Route length
- Right of way conditions
- A rough quantity survey and a general cost analysis are present.

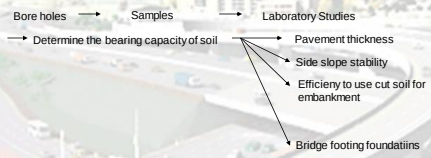
HIGHWAY ROUTE STUDY

Survey: A more detailed research on the feasible routes selected after the preliminary survey phase is followed on the survey phase. In this phase, topographical survey and soil surveys are made in general.

A) Topographical Survey: Topographical maps with higher scales have to be produced about the route alternatives remaining after the preliminary survey.
The routes drawn at preliminary survey phase are transferred to the new maps drawn after the necessary corrections are made. The plan, the profile and the cross-sections are prepared for each route alternative. After this phase, a preliminary project is obtained. On the maps prepared, **engineering structures**, **intersections** with the other highways, **material suppliance places**, facilities of the highway and **private properties** have to be shown.

HIGHWAY ROUTE STUDY

B. Soil Survey: Soil surveys occupy an important role on the survey phase. Detailed geological and geotechnical surveys are prepared for each route alternative. Bore holes are opened on specific places and with specific distances between them during these surveys. The samples obtained are sent to laboratories to be researched. Underground water level also has to be researched



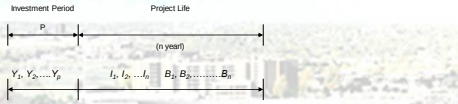
HIGHWAY ROUTE STUDY

Economical comparison: The last comparison made to determine the certain route. Using the data gathered, information about the various highway cost items are obtained. It is possible to find the most efficient route alternative with the economical comparison in the light of them.



HIGHWAY ROUTE STUDY

For the methods listed above, equivalent amounts on highways investments and the other cost and benefits of the highway are calculated.



Discount process expresses the inverse of the interest process. Actuilization of (Y) to the future year with the interest rate of *i*.

$$Y = Y_1 * (1+i)^p + Y_2 * (1+i)^{p-1} + \ldots\ldots + Y_p * (1+i)^1$$

Actuilization of (I and B) to the present years with the interest rate of *i*.

$$I = \frac{I_1}{(1+i)^1} + \frac{I_2}{(1+i)^2} + \ldots\ldots + \frac{I_{n-1}}{(1+i)^{n-1}} + \frac{I_n}{(1+i)^n}$$

$$B = \frac{B_1}{(1+i)^1} + \frac{B_2}{(1+i)^2} + \ldots\ldots + \frac{B_{n-1}}{(1+i)^{n-1}} + \frac{B_n}{(1+i)^n}$$

HIGHWAY ROUTE STUDY

Selection of the routes by using benefit cost ratio.

Benefit: Reduction on operation cost (ΔI)
Cost: Investment + Maintenance/Rehabilitaion (Y+B)

$$benefit/Cost = \frac{\Delta I}{Y + B}$$

If there is not any exiting route;

Route 1 B/C= $\frac{I_1}{Y_1 + B_1}$ and Route 2 B/C= $\frac{I_2}{Y_2 + B_2}$

If there is an exiting route;

Route 1 B/C= $\frac{I_{existing} - I_1}{Y_1 + B_1}$ and Route 2 B/C= $\frac{I_{existing} - I_2}{Y_2 + B_2}$